

Lab 1 – Requirements Table

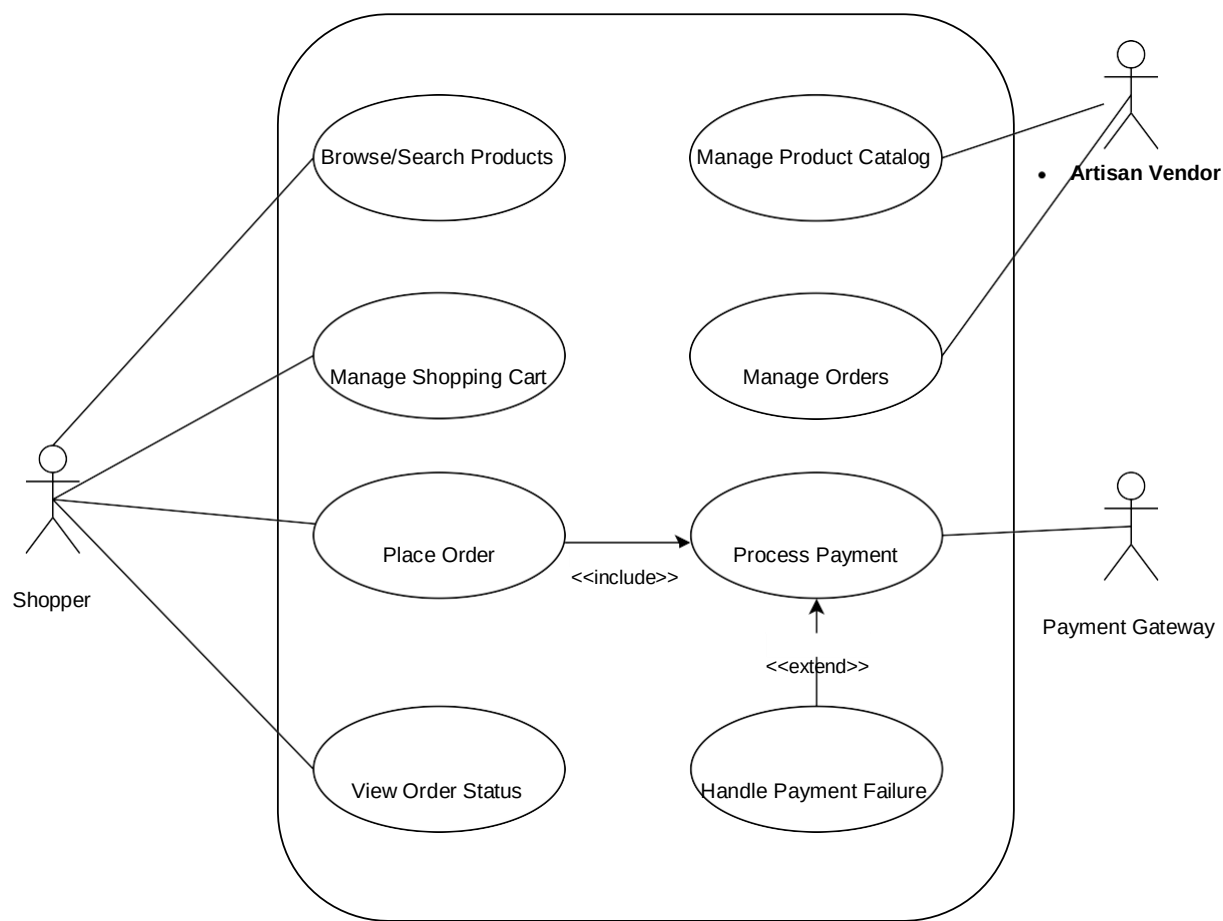
Problem Statement #31: Multi-Vendor Artisan E-Commerce Marketplace

GITHUB LINK :- <https://github.com/Darshil085/SE--Labs--PES1UG25CS820-->

Req Id	Type	Description	Priority	Acceptance criteria	Relationale
FR-001	Functional	The system shall split customer cart payments at checkout, allocating respective item earnings to multiple independent vendor accounts after deducting a 5% platform fee.	High	Pass: Split payout calculations correctly balance to the total cart value after deducting the 5% platform fee. Fail: Any discrepancy occurs in the payout calculation.	Ensures correct payment distribution among multiple artisan vendors and the platform.
FR-002	Functional	The system shall allow Artisan Vendors to add, update, and remove products from their product catalog.	High	A vendor can successfully add a product with its name, description, price, and available quantity, update its details, or remove it from the catalog.	Allows vendors to manage the products they sell.
FR-003	Functional	The system shall allow Shoppers to browse, search, and view products offered by multiple Artisan Vendors.	High	When a shopper searches for a valid product or selects a product category, matching products and their details are displayed.	Allows shoppers to discover and select products for purchase.
FR-004	Functional	The system shall allow Shoppers to add products from multiple Artisan Vendors to a single shopping cart and proceed to checkout.	High	A shopper can add products from at least two different vendors to one cart, modify quantities or remove items, and successfully proceed to checkout.	Supports the core multi-vendor shopping experience.
FR-005	Functional	The system shall allow Artisan Vendors to view received orders and update the status of their orders.	High	A vendor can view orders containing their products and update the order status to a valid status such as Processing, Shipped, or Completed.	Allows vendors to manage and fulfil customer orders.

NFR-001	Non-Functional	The product catalog shall support high-resolution image rendering with CDN caching, delivering load times of less than 500 ms.	High	Pass: Benchmarking tests confirm product images load in less than 500 ms under simulated peak load. Fail: The load time exceeds 500 ms.	Ensures fast product browsing and a smooth user experience
NFR-002	Non-Functional	The system shall ensure that only authenticated users can access protected shopper and Artisan Vendor account functions.	High	Pass: Unauthenticated users are denied access to protected account pages and authenticated users can access only their authorized functions. Fail: An unauthorized user can access protected account functions.	Protects user accounts and sensitive marketplace information.

Multi-Vendor Artisan E-Commerce Marketplace



Multi-Vendor Artisan E-Commerce Marketplace

Use-Case Flow Specification

Use Case: Place Order

Preconditions

- The shopper has selected one or more products.
- The selected products are available in the marketplace.
- The shopper has added the products to the shopping cart.

Postconditions

- The order is successfully created.
- Payment is processed successfully.
- The respective Artisan Vendor receives the order details.
- The system deducts a 5% platform fee and allocates the remaining earnings to the respective vendor accounts.

Main Success Scenario

1. The shopper browses and selects products.
2. The shopper adds the selected products to the shopping cart.
3. The shopper proceeds to checkout.
4. The system displays the order summary and total amount.
5. The shopper confirms the order and selects a payment method.
6. The system sends the payment request to the Payment Gateway.
7. The Payment Gateway successfully processes the payment.
8. The system creates the order.
9. The system deducts the 5% platform fee and calculates the respective vendor earnings.
10. The system sends the order details to the respective Artisan Vendor.
11. The system displays an order confirmation to the shopper.

Alternate Flow: Payment Failure

1. The Payment Gateway reports that the payment has failed.
2. The system informs the shopper that the payment was unsuccessful.
3. The shopper may select another payment method and try again.
4. If payment is successful, the main success scenario continues from Step 7.
5. If the shopper cancels or payment continues to fail, the order is not created.